

Project Design Phase

Problem – Solution Fit Template

Date	2 July 2026
Team ID	
Project Name	Nutrition Assistant Using MERN stack
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem–Solution Fit for this project focuses on identifying the challenges faced by individuals in maintaining a healthy lifestyle, tracking daily nutrition, and planning balanced meals. The proposed Nutrition Assistant, developed using the MERN Stack, provides an intelligent and user-friendly platform that analyzes user health data and delivers personalized nutrition recommendations, enabling healthier eating habits and informed dietary decisions.

Purpose:

- To simplify daily nutrition management through an interactive and easy-to-use web application.
- To help users maintain a balanced diet by providing personalized meal plans based on their health profile, BMI, calorie requirements, and dietary preferences.
- To enable users to track their daily calorie intake, nutritional values, water consumption, and overall health progress.
- To provide a centralized platform for monitoring health metrics, analyzing
- Nutrition trends, and receiving AI-based food recommendations through intuitive dashboards and visual insights.
- To improve awareness of healthy eating habits, encourage evidence-based dietary decisions, and support long-term health and wellness through continuous progress tracking.

Template:

Problem-Solution fit canvas 2.0

Purpose / Vision: Enable individuals to achieve healthier lifestyles by providing personalized nutrition guidance, AI-powered meal recommendations, and nutrition tracking through an interactive MERN Stack web application.

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS - Health-conscious individuals - Fitness enthusiasts - Students and working professionals - People managing weight (loss/gain) - Individuals with lifestyle diseases (obesity, diabetes, hypertension) - Dieticians and Nutritionists - Gym trainers and wellness coaches	6. CUSTOMER CONSTRAINTS CC - Lack of nutritional knowledge - Busy lifestyle and limited time - Difficulty maintaining consistency - Conflicting and confusing information online - Limited personalized guidance - Manual tracking is time-consuming	5. AVAILABLE SOLUTIONS AS - Manual calorie calculation - Diet charts from nutritionists - Generic fitness applications - Health websites and blogs - Food tracking mobile apps - YouTube fitness and diet videos	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P - Track daily calorie and nutrient intake - Calculate BMI and daily calorie requirements - Plan balanced meals - Receive personalized nutrition recommendations - Monitor health progress over time - Improve eating habits - Achieve fitness and wellness goals			
Identify strong TR & EM	3. TRIGGERS TR - Desire to lose or gain weight - Doctor or dietician recommendations - Beginning a fitness journey - Health awareness campaigns - Lifestyle-related health concerns - Need for better meal planning	10. YOUR SOLUTION SL A Nutrition Assistant using MERN Stack that provides: - Secure User Authentication - Personalized User Profile - BMI Calculator - Calorie Requirement Calculator - Daily Nutrition Tracking - AI-powered Meal Recommendations - Personalized Meal Planner - Nutrition Analysis Dashboard - Progress Tracking and Reports - Food Database Management - Responsive Web Interface - Admin Panel for managing users and food data	8. CHANNELS OF BEHAVIOUR CH Online - Nutrition Assistant Web Application - Social Media (Instagram, YouTube) - Health blogs - Fitness communities - Google Search - Email notifications Offline - Hospitals and clinics - Dieticians - Gyms and fitness centers - Wellness workshops - Family and friends - Health camps	Extract online & offline CH into BE
	4. EMOTIONS: BEFORE / AFTER EM Before: - Confused about healthy food choices - Frustrated with calorie calculations - Lack of motivation - Difficulty maintaining a balanced diet - Uncertain about nutritional requirements After: - Confident in food choices - Motivated to maintain healthy habits - Better understanding of nutrition - Improved health awareness - Satisfaction from achieving fitness goals			

CS Who are your customers?
 J&P What jobs do they want done?
 TR What triggers them?
 EM How do they feel?
 CC What holds them back?
 RC Why is it a real problem?
 BE What do they do?
 CH Where do they do it?
 SL How will you solve it?
 AS What solutions exist?